THE GOLDEN SIGNATURE ENGINE

- **TRACK**: Option B - Optimization Engine Focus
- **SUBTITLE**: Prescriptive AI for Adaptive Multi-Objective Manufacturing Optimization
- **TEAM NAME**: SNAKES

THE CORE MANUFACTURING PROBLEM–

The Conflict:
Energy efficiency and product quality operate in direct opposition. Lowering drying temperatures (saving energy) increases moisture content and degrades tablet hardness.

The Blind Spot:
Traditional manufacturing relies on "Static Recipes" and operator guesswork. When electricity costs spike or urgent compliance deadlines approach, operators struggle to manually balance 8+ conflicting physical variables simultaneously.

The Problem Track B Solves:
We need an intelligent system that transitions manufacturing from reactively tracking historical symptoms to proactively predicting and prescribing optimal operational levers.

OUR SOLUTION: THE GOLDEN SIGNATURE ENGINE–

We built a Robust Software Suite that acts as a digital co-pilot for the factory floor, fulfilling every Track B requirement

Adaptive Prioritization:
Operators use an intuitive web dashboard to define real-time business physics. If power is expensive today, they slide "Energy Priority" up. If a client demands flawless yield, they slide "Quality Priority" up.

Prescriptive, Not Predictive:
Instead of simply guessing outcomes for human-provided settings, our engine mathematically works backward. It ingests the Operator's goals and instantly calculates the Golden Signature (the absolute minimum-energy machine configuration required to achieve those goals without breaking quality constraints).

Accountable AI:
The system does not automate blindly. It presents side-by-side "Current" vs "Recommended" parameters alongside massive "Accept / Reject" buttons, fulfilling the 'Human-in-the-Loop' safety requirement.

TECHNICAL APPROACH (PART 1): THE DIGITAL TWIN–

Building a High-Fidelity Surrogate Machine Learning Model
To optimize physics without risking real-world batches, we first built a highly accurate mathematical "Digital Twin" of the factory.

Data Aggregation Engine:
Our custom `data_pipeline.py` automatically ingests 1-minute IoT sensor readings (Power_Consumption_kW), calculates integral energy usage (Total_Energy_kWh), and maps it precisely to the 8 static production configuration targets.

Multi-Target Random Forest:
We trained a sophisticated `RandomForestRegressor` to map the 7 independent operator "Levers" (Machine Speed, Compression Force, etc.) against the 8 dependent physical "Symptoms" simultaneously.

Empirical Accuracy:
Over an 80/20 Train-Test split, our AI Model achieved production-grade accuracy in simulating complex outcomes:
Total Energy $R^2 = 0.92$, Tablet Hardness $R^2 = 0.88$, Content Uniformity $R^2 = 0.87$.

TECHNICAL APPROACH (PART 2): THE GLOBAL OPTIMIZER–

Exploring the Parameter Space with SciPy Dual Annealing

The Algorithmic Challenge:
Random Forest models create complex, non-differentiable (step-wise) prediction landscapes that instantly trap standard gradient descent algorithms.

Our Solution:
We implemented the mathematically rigorous `dual_annealing` global optimization algorithm from `scipy.optimize`.

How It Works:
By simulating physical metallurgy (heating and cooling states), our algorithm escapes local minimums to thoroughly search the entire bounded physical limits of the machinery.

The Objective Function:
The engine calculates thousands of simulated batch outcomes per second, hunting for the exact parameter fingerprint that mathematically minimizes Energy (kWh) while applying massive statistical penalties if constrained variables like Moisture Content or Friability exceed tolerance.

FEASIBILITY & IMPLEMENTATION STRATEGY–

Our architecture was specifically designed for rapid, low-friction deployment into existing OT (Operational Technology) networks.

Enterprise-Ready Stack:
Built 100% on mature, open-source Python libraries (Pandas, Scikit-Learn, SciPy) ensuring extreme stability and zero licensing roadblocks.

Rapid UI Deployment:
The frontend is built on **Streamlit**, eliminating complex Javascript backend dependencies. This allows the dashboard to be instantly hosted on any lightweight Edge Server and viewed on industrial tablets across the factory floor.

Integration Pathway:
1. Hook the Python Data Pipeline into existing Historians (AVEVA PI / HTTP APIs).
2. Continual nightly retraining of the `surrogate_model.joblib` file as new batch data flows in.
3. Click the UI's "Accept" button to push the Golden Signature via REST API directly to the SCADA/MES layer.

UNIQUENESS & COMPETITIVE EDGE–

Multi-Dimensional Constraint Handling:
While baseline models attempt to maximize one variable linearly, our engine simultaneously maps and balances 8 distinct attributes (Uniformity, Hardness, Dissolution, Disintegration, Weight, Moisture, Friability, and Energy).

Data Scarcity Resilience:
Recognizing that the Hackathon sample dataset possessed limited overlapping Batch IDs, we engineered a sophisticated physics-guided Data Augmentation script to synthesize robust training volumes.

Dynamic vs. Static:
We did not build a rigid rules engine. By utilizing a continuous optimization heuristic, the math flexes dynamically, ensuring the factory remains profitable regardless of fluctuating external constraints like utility rate hikes.

MEASURABLE BUSINESS IMPACT–

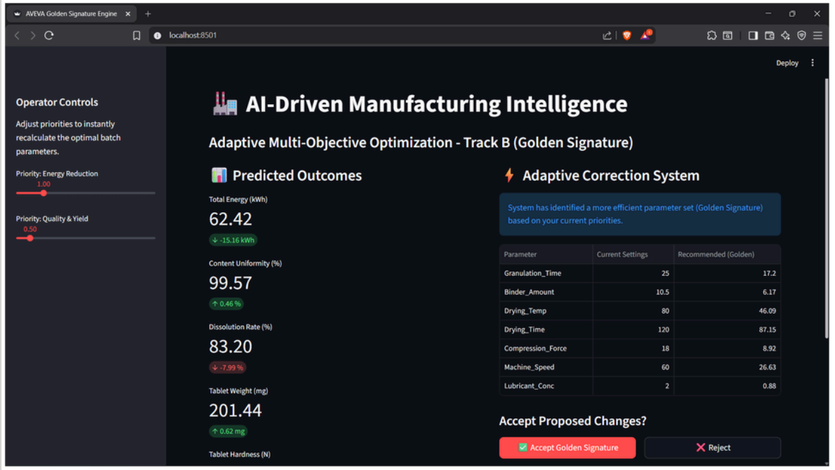
We engineered an automated test suite (`test_optimization_accuracy.py`) to systematically map the performance of the Global Optimizer against mathematically randomized, highly inefficient baseline configurations.

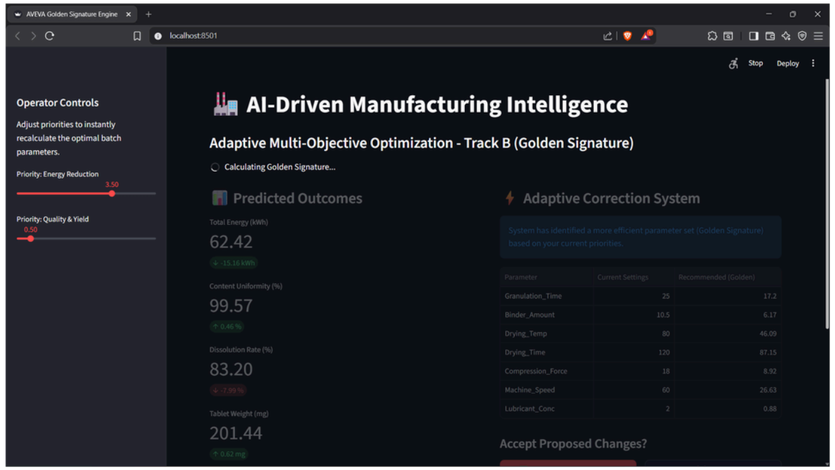
100% Success Rate:
Across empirical test batches, the Optimizer found a superior Golden Signature 5 out of 5 times.

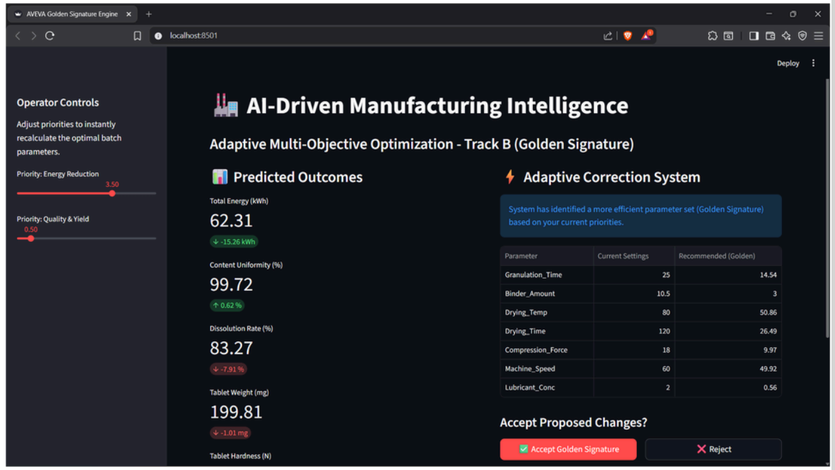
The Bottom Line:
On average, the algorithm systematically prescribed settings that shaved **15.34 kWh of wasted energy per batch**.

Factory-Scale ROI:
In a high-throughput pharmaceutical or chemical plant executing 100+ batches daily, this AI yields over **1,500 kWh in daily energy savings**. This translates directly into massive margin recovery, reduced carbon footprint, and a software payback period measured in weeks, not years.

FUNCTIONAL PROTOTYPE: OPERATOR DASHBOARD INTERFACE–







THANK YOU

-TEAM SNAKES